

AKBARJON ROZIMOV

Ph.D. Candidate — Evolutionary & Conservation Genomics

CONTACT

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Current Affiliation:
Institute of Zoology, CAS
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Nationality: Uzbekistan
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RESEARCH IDs

› **ORCID:** 0000-0001-5628-1524
› **RG:** profile/Akbarjon-Rozimov
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SKILLS

Technical
› Seine netting for fishing, paddling
› Driving (license B)

Taxonomy & Ecology
› Species description, Check listing
› Fish growth & stock assessment
› IUCN & Regional Red List assessment

Genomics
› Mitochondrial & Chromosome-level Genome Assembly
› Annotation, SNP calling & Genotyping

Population Genomics
› PCA, ADMIXTURE, Phylogenomics
› f3- and D-statistics, Gene flow

Demographic & Conservation
› N_e trajectories, Divergence time
› Inbreeding, Genetic load

SOFTWARE

GATK · PLINK · VCFtools · BCFtools · bedtools · ADMIXTURE · Dsuite · WinPCA · Saguaro · fastsimcoal2 · PSMC · TreeMix · SLiM · SMC++ · Stairway-plot 2 · ImageJ · Adobe Illustrator · GetOrganelle · ArcGIS and many more

Bash · R · Python · HPC workflows

COURSES & CERTIFICATES

RESEARCH PROFILE

Evolutionary biologist and ichthyologist specializing in the genomics and conservation of Central Asian freshwater fishes. Ph.D. research delivers the first chromosome-level genomic resources and a conservation assessment for the critically endangered Amu Darya false shovelnose sturgeons (*Pseudoscaphirhynchus kaufmanni* and *P. hermanni*), integrating population genomics, demographic inference, and conservation genomics analyses with field-based biodiversity surveys. Six peer-reviewed publications (*npj Biodiversity*, *Oryx*, *Zoosystematics and Evolution*), IUCN SSC Sturgeon Specialist Group member, and peer reviewer for several journals. Seeking a postdoctoral position applying quantitative and simulation modelling in ecology and genomics.

EDUCATION

Ph.D. in Evolutionary Genomics — **September 2023 – Present**
Institute of Zoology, Chinese Academy of Sciences — Beijing, P.R. China
Fish Evolution and Genomics Group (Prof. Baocheng Guo) — State Key Laboratory of Animal Biodiversity Conservation and Integrated Pest Management

- Thesis: Conservation Genomics of the Amu Darya False Shovelnose Sturgeons.
- Supervisors: Prof. Baocheng Guo and Dr. Xinxin Li.
- Speciation, demography and conservation genomics of Amu Darya shovelnoses utilizing high-coverage chromosome-level genomic data.

M.Sc. in Ichthyology and Hydrobiology — **September 2021 – June 2023**
National University of Uzbekistan named after Mirzo Ulugbek — Tashkent

- Thesis: The ichthyofauna of the Chirchik River — species diversity, conservation and practical importance. Supervisors: Prof. Matnazar Rakhimov and Dr. Bakhtiyor Sheraliev.

B.Sc. in Biology — **September 2017 – June 2021**
Urgench State University, Faculty of Life Sciences — Khorezm, Uzbekistan

- Undergraduate research: Diversity and conservation of freshwater fishes in the lower Amu Darya River. Supervisor: Dr. Bakhtiyor Sheraliev.

PUBLICATIONS

Peer-reviewed articles, manuscripts, most recent first.

- **Rozimov A**, **Zhang H**, **Fan L**, Li X, … Guo B. Genomic insights into the speciation and conservation of the Amu Darya False Shovelnose Sturgeons. *(in preparation)*
- Karimov E, Karimov B, Care J, Omonov O, **Rozimov A**, Azizov F, Osmonov I, Konurbayev E, Freyhof J, De Keyser J, Schletterer M, Hayes D. Distribution of fish species in rivers of Central Asia: an update on current biogeography. *(in preparation)*
- **Rozimov A**, Li X, Karimov E, Wang C, Guo B, Freyhof J. Taxonomy of snow barbels of the genus *Schizothorax* in the Aral Sea basin (Teleostei, Cyprinidae). *(in preparation)*
- Li X, Zhang H, Fan L, **Rozimov A**, Kholmatov B, Yang J, … Guo B (2026) Initiation of the Chondrostei Genome Project (CGP). *Aquatic Diversity and Ecology* 100010. doi:10.1016/j.ade.2026.100010
- **Rozimov A**, Li X, Sheraliev B, et al. (2026) Multidimensional crisis in the conservation of Amu Darya false shovelnose sturgeons. *npj Biodiversity*. doi:10.1038/s44185-026-00141-w
- **Rozimov A**, Turaev L, Namozov S, Agzamov F, Sobirov J, Wang C, Kholmatov B, Li X, Guo B (2025) Implications for habitat suitability using length–weight relationships and condition factors in the Chirchik River, Syr Darya Basin. *Journal of Applied Ichthyology* 2025: 6648940, 10 pp. doi:10.1155/jai/6648940
- **Rozimov A**, Wang Y, Wang M, Zou M, Sobirov J, Karimov E, Kholmatov B, Freyhof J, Namozov S, Wang C, Li X, Guo B (2025) Mitochondrial genome insights into the phylogenetics and biogeographic evolution of snow trout (Cyprinidae, Schizothorax) in the Tien Shan Mountains. *Zoosystematics and Evolution* 101(1): 91–102. doi:10.3897/zse.101.137787
- Sheraliev B, Kayumova Y, Allayarov S, **Rozimov A**, Komilova D, Urmonova D, Peng Z (2022) Length–weight relations of 14 endemic and indigenous freshwater fish species (Actinopterygii) from the Aral Sea basin, Uzbekistan. *Acta Ichthyologica et Piscatoria* 52(4): 239–243. doi:10.3897/aiep.52.89279
- Sheraliev B, **Rozimov A**, Ludwig A, Peng Z (2021) Rediscovery of rare shovelnose sturgeons in the Amu Darya River, Uzbekistan. *Oryx* 55(3): 332. doi:10.1017/S0030605321000211

CONFERENCE PRESENTATIONS

10th International Symposium on Sturgeon — Yichang, China — **Oct 2025**

- Oral presentation: “Multidimensional crisis: geographic range contraction, phenotypic degradation, genomic erosion, and cultural exploitation of the critically endangered Amu Darya shovelnose sturgeons.”

› Finding Hidden Messages in DNA (Bioinformatics I) – UC San Diego / Coursera, October 2026
› Genome Sequencing (Bioinformatics II) – UC San Diego / Coursera, October 2026
› Comparing Genes, Proteins, and Genomes (Bioinformatics III) – UC San Diego / Coursera, October 2026
› Command Line in Linux / Coursera, October 2026
› Molecular Evolution (Bioinformatics IV) – UC San Diego / Coursera, October 2026
› Genomic Data Science and Clustering (Bioinformatics V) – UC San Diego / Coursera, October 2026

LANGUAGES

› Uzbek — Native
› English — B2 / C1 (writing)
› Russian — Working proficiency
› Chinese — HSK 2

MEMBERSHIPS

› IUCN SSC Sturgeon Specialist Group (2021–present)
› World Sturgeon Conservation Society (2025–present)

AWARDS

› ANSO Scholarship for Young Talents 2023 — No. 2023ANP0152, UCAS
› Presidential Scholarship 2022/2023 — No. 4/1, NUUz

HOBBIES

› Bodybuilding and jogging

30th Anniversary Meeting, IUCN Sturgeon Specialist Group — Bordeaux, France Sept 2024

- Invited online presentation: "Sturgeon survival and conservation challenges in the Amu Darya."

RESEARCH EXPERIENCE & FIELDWORK

Curator of the Zoological Collection January 2022 – August 2023

Institute of Zoology, Academy of Sciences of the Republic of Uzbekistan — Tashkent

- Managed inventory, curation, and digitization of the national ichthyological collection; conducted data analysis supporting taxonomic and conservation research.

Selected field campaigns

- Lost Syr Darya Sturgeon survey (2025) — trammel-net and eDNA surveys across the Syr Darya basin in Uzbekistan, Kazakhstan and Kyrgyzstan; with Tennessee Aquarium and Re:wild.
- Sturgeon and cyprinid sampling for de novo genome assembly (2024) — *Pseudoscaphirhynchus kaufmanni*, *Acipenser gueldenstaedtii*, *Schizothorax eurycephalus*, *Gymnodiptychus kessleri*.
- Hydro4U fish sampling (2022–2023) — Kyrgyzstan and Uzbekistan, with TIIAME.
- Chirchik, Zeravshan, Surkhandarya and lower Amu Darya river surveys (2019–2023) — partly self-funded inventories underpinning three publications.

PROFESSIONAL SERVICE

Peer reviewer (2023–present): *Central Asian Journal of Water Research*; *Journal of Ichthyology*; *Zoosystematics and Evolution*; *Ecologica Montenegrina*; *ZooKeys*; *Frontiers in Aquaculture*; *Journal of Applied Ichthyology; Water Biology and Security*

REFERENCES

1. Prof. Baocheng Guo – Principal Investigator, Fish Evolution and Genomics Group
Institute of Zoology, Chinese Academy of Sciences — guobaocheng@ioz.ac.cn

2. Dr. Xinxin Li – Associate Professor, Fish Evolution and Genomics Group
Institute of Zoology, Chinese Academy of Sciences — lixinxin@ioz.ac.cn